silhouette scores in separating these clusters, indicating they are most effective at distinguishing between query types. This finding aligns with prior research (Li et al., 2024; Cao et al., 2024), which has identified middle layers as crucial for safety-related behaviors. We designate the layer with the highest silhouette score as our *target layer*, denoted by l^*.

Computing the Refusal Vector. Having identified the target layer, we extract the refusal vector using the difference-in-means approach (Zou et al., 2023a; Arditi et al., 2024). Specifically, we compute the refusal vector as:

$$R := \frac{1}{|Q^-|} \sum_{q^- \in Q^-} a^{l^*}(q^-) - \frac{1}{|Q^+|} \sum_{q^+ \in Q^+} a^{l^*}(q^+) \quad (1)$$

where $a^{l*}(\cdot)$ represents the activations at the target layer. This vector captures the directional shift from response-generating to refusal-generating representations.

3.2. Setting Alignment Targets

Given the extracted refusal vector, the next key question is how to set appropriate targets for shifting the model's representations during fine-tuning.

Would Uniform Shifts Work? A straightforward idea, adopted by previous inference-time methods (Cao et al., 2024; Wang et al., 2024), would be to apply *uniform shifts*: these methods directly add or subtract the refusal vector from activations by a fixed amount—adding it to increase refusal likelihood for harmful queries and subtracting it to reduce refusal for safe queries. However, when adapting this uniform shift idea to our fine-tuning setting, our empirical analysis reveals that it fails to effectively balance safety and helpfulness, ultimately leading to model breakdown where the model fails to maintain coherent responses across query types (Appendix B).

Analysis of the embedding space geometry, as shown in Figure 3 (Left), reveals why uniform shifts are inadequate: different queries require different magnitudes of shift to alter the model's behavior. Over-refused queries that lie farther from the safe cluster require larger shifts, while those closer to it need smaller adjustments. We hypothesize that model breakdown occurs because forcing all activations to meet the same large target shift makes it difficult for the model to reconcile competing objectives—maintaining coherent responses while adjusting its refusal behavior.

Towards Individualized Shifts. The failure of uniform shifts motivates our approach of individualized shifts, which calibrates activation targets for each query based on its specific location in the embedding space. Rather than forcing all queries to meet the same shift target, we tune each query's representation by the amount *necessary* to correct over-refusal behaviors. This "*just enough*" shift reduces the risk of imposing excessively large changes on pseudo-harmful queries that are already near the safe cluster, thereby allowing the model to adjust more flexibly to maintain performance on rejecting genuinely harmful queries. In other words, a uniform shift could overconstrain the model and overshadow other goals, whereas a just-enough activation target provides a softer requirement, allowing the model's parameters to adjust more readily for other objectives.

What is the "Just Enough" Shift? An Empirical Angle. To establish the ground truth of the just enough shift, we need to understand how much shift along the refusal vector is required for each pseudo-harmful query to change the model's response from refusal to acceptance. To simplify the notation, we will use a_q to denote the embedding $a^{l^*}(q)$ extracted from query q at the target layer. For each over-refused query q, we compute:

$$a_q - \gamma R \quad (2)$$

where R is the refusal vector and γ characterizes the shift magnitude. To find the just-enough shift, we perform a line search for γ in the range $[0.1, 1]$ with a step size of 0.1 to find the minimal value that generates a compliant response for each query, denoted as γ_q^*. We use GPT-4o as a judge to determine whether each response is compliant or a refusal (Appendix E).

However, using these ground truth just enough shifts as targets during fine-tuning would be computationally expensive due to the search, generation, and judgment processes involved. This motivates us to find an efficient proxy for the just enough shift. Given that we are working in the embedding space, a natural proxy to examine is how much of the query's representation contributes to refusal behavior. Intuitively, if a query's representation contains stronger refusal components, it should require more correction to induce acceptance. We can measure this refusal contribution through the query's projection onto the refusal vector:

$$\mathrm{Proj}_R(a_q) \;=\; \frac{R \cdot a_q}{\|R\|^2}\, R. \quad (3)$$

A larger projection indicates stronger alignment with the refusal direction.

Remarkably, we found a strong linear relationship between the necessary shift γ_q^* and the magnitude of the projection, $\|\mathrm{Proj}_R(a_q)\|_2$, as illustrated in Figure 3 (Right). This linear relationship has important implications for our method. First, it validates our geometric intuition that queries more strongly aligned with the refusal direction require proportionally larger shifts to overcome refusal behavior. More importantly, it provides us with a computationally efficient way to estimate the necessary shift magnitude: instead of performing expensive search and generation processes to

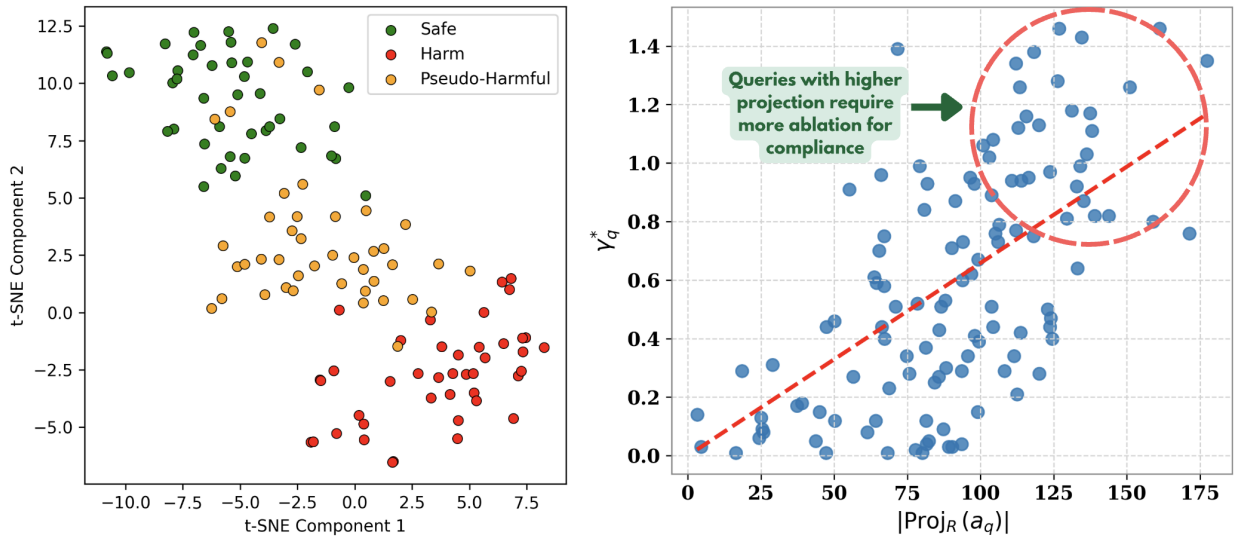

Figure 3. **(Left)** Last token query activations of Pseudo-Harmful Queries occupy the activation space between safe and harmful clusters. This highlights the importance of individualistic treatment of queries to mitigate over-refusals. Activations are extracted from layer 13 of the Llama-2-7b-chat **(Right)** This figure visualizes the relationship between the projection magnitude onto the *Refusal Direction* and the minimal scaling factor, γ, that must be removed to generate a compliant response. A higher projection magnitude indicates a greater need for adjustment (γ) to achieve compliance. The Pearson correlation coefficient between the projection magnitude and γ is 0.63, highlighting a positive correlation between refusal intensity and the required correction.

find γ_q^* for each q, we can simply compute the projection magnitude for q and scale it properly with a shared constant. In other words, the necessary shift is the refusal vector scaled by an amount proportional to the projection magnitude. Since the projection itself is in the direction of the refusal vector, this suggests a simple form for our target activations:

$$a_q - \alpha \, \text{Proj}_R(a_q) \tag{4}$$

where α is a constant.

Deriving the "Just Enough" Shift with a Simplified Theoretical Model. To further build intuition on why projection-based shifts are effective, let us analyze a simplified model of the refusal mechanism in the embedding space. Consider a *locally* linear approximation of the refusal boundary: $R \cdot a_q = d$, where d is some threshold. Under this model, the model refuses queries where $R \cdot a_q > d$ and accepts those where $R \cdot a_q \leq d$. This linear boundary approximation is motivated by our earlier observation that harmful and benign queries cluster in different regions of the embedding space, with R capturing the principal direction separating these clusters.

Given this model, an over-refused query is one where $R \cdot a_q > d$ despite being benign. To correct such a query, we need to shift its activation a_q by some Δ_a such that the shifted activation $\tilde{a}_q = a_q + \Delta a$ lies exactly on the boundary. By geometric reasoning, the minimal shift Δa that brings a_q to the boundary must be parallel to R, as this represents the shortest path to a hyperplane. Therefore, we can write $\Delta_a = \beta R$ for some scalar β. To find β, we require that the shifted activation satisfies the boundary equation:

$$R \cdot (a_q + \beta R) = d \tag{5}$$

Solving this equation yields:

$$\beta = \frac{d - (R \cdot a_q)}{\|R\|^2} \tag{6}$$

Thus, the minimal required shift is:

$$\Delta_a = \frac{d - (R \cdot a_q)}{\|R\|^2} R = \left(\frac{d - R \cdot a_q}{R \cdot a_q} \right) \text{Proj}_R(a_q) \tag{7}$$

For over-refused queries that lie close to the decision boundary, $R \cdot a_q$ is near d. In this case, the numerator $(d - R \cdot a_q)$ in the multiplier becomes small while the denominator $R \cdot a_q$ remains stable near d. Therefore, the multiplier itself becomes small and approximately constant, supporting our empirically-motivated approach of using scaled projections.

Remark. Our experiments demonstrated that we can find a linear hyperplane in the embedding space that perfectly separates rejected queries from answered ones, lending support to the linearity assumption underlying our analysis. However, fully verifying this assumption is intractable due

to the impossibility of comprehensively sampling the high-dimensional embedding space. We emphasize that this theoretical analysis primarily serves to build intuition for our proposed alignment targets rather than providing a rigorous justification for their design.

3.3. Overall Algorithm Design

Building upon the discussion above, we propose the following *Projection-Calibrated Refusal Direction Loss* to fine-tune a safety-aligned model to correct over-refusal behaviors:

$$\mathcal{L}_{\text{PRD}}(\theta, q) \;=\; 1 - \cos_\text{sim}\Big(a_\theta^{l*}(q), a^{\text{tgt}}(q)\Big) \tag{8}$$

where $\cos_\text{sim}$ denotes cosine similarity. This loss encourages the model to shift $a_\theta^{l*}(q)$ toward $a^{\text{tgt}}(q)$, where $a^{\text{tgt}}(q)$ is defined differently for each query type:

$$a^{\text{tgt}}(q) = \begin{cases} a_q - \alpha\,\text{Proj}_R(a_q) & \text{if } q \in Q^{PH} \cup Q^{+} \\ a_q + \alpha\,\text{Proj}_R(a_q) & \text{if } q \in Q^{-} \end{cases} \tag{9}$$

Here, Q^{PH} is the set of pseudo-harmful queries, Q^{+} is the set of safe queries, and Q^{-} is the set of genuinely harmful queries. For safe and pseudo-harmful queries, the target activations guide the model away from the refusal direction to reduce the refusal tendency, while for harmful queries, the targets shift representations along the refusal vector to strengthen the model's defensive mechanisms. The target activations are computed using the model's activations before each update step. α is hyperparameter.

The overall workflow of our algorithm is iterative, alternating between two key phases: refusal vector identification and fine-tuning. In the refusal vector identification phase, we compute the refusal vector R using the current model's activations. During fine-tuning, we iteratively update the model's parameters to align its activations with query-specific targets $a^{\text{tgt}}(q)$. For each query, we calculate the target activations based on its type (safe, pseudo-harmful, or harmful) from the current model, then perform gradient updates to minimize the Projection-Calibrated Refusal Vector Loss $\mathcal{L}_{\text{PRD}}(\theta, q)$. The pseudo-code for ACTOR can be found in Appendix A.

Our approach offers significant advantages over traditional instruction tuning in terms of computational efficiency. First, our fine-tuning process relies only on query activations, avoiding the need for costly output generation. Second, as the model only requires updating parameters before the target layer l^*, rather than the entire network.

4. Experiments

4.1. Experimental Settings

Refusal Direction Calculation. We compute the refusal direction for target layers using harmful queries from HexPhi (Qi et al., 2023) and benign queries from TruthfulQA (Lin et al., 2021). Specifically, we randomly select 64 harmful and 64 benign queries to extract these directions, as detailed in Section 3.1.

Training Datasets. To train the models, we constructed a dataset $\mathcal{D}_{\text{train}} = \{\mathcal{D}_{\text{harmful}}, \mathcal{D}_{\text{benign}}, \mathcal{D}_{\text{pseudo}}\}$, comprising of harmful, benign, and pseudo-harmful queries. Harmful prompts are taken entirely from the HexPhi benchmark (Qi et al., 2023). Benign prompts consist of 210 items randomly selected from the UltraChat dataset (Bartolome et al., 2023), obtained by sampling 30 random examples from each of the dataset's seven topical categories to ensure broad coverage of benign user intents. Pseudo-harmful prompts are compiled by drawing 25 random items from each of XSTest (Röttger et al., 2023), SCOPE (Zeng et al., 2024), OR-Bench-Hard-1K (Cui et al., 2024), and PHTest (An et al., 2024). Reference answers for all harmful and pseudo-harmful prompts are generated using GPT-4o (Hurst et al., 2024). Ablation studies that vary both the number of benign UltraChat examples (n = 15, 50) and the number of over-refusal prompts (n = 10, 50) are reported in Appendix C.3 and C.4.

Evaluation Datasets. We evaluate over-refusal using the held-out versions of XSTest (150 samples), SCOPE (593 samples), OR-Bench-Hard-1K (1219 samples), and PHTest (1977 samples) benchmarks with OKTest (Shi et al., 2024) being used as an Out-Of-Distribution (OOD) benchmark. We also include comprehensive ablation studies—where the model is trained on subsets of the above datasets (e.g., only XSTest or only OR-Bench) and evaluated on the remaining held-out or unseen benchmarks—to probe out-of-domain generalization; these results are reported in Appendix C.5. Beyond mitigating exaggerated safety, maintaining the safety on genuinely harmful queries is crucial. To assess safety after fine-tuning with ACTOR, we utilize AdvBench (Zou et al., 2023b). Following SCANS (Cao et al., 2024), we evaluate the general model capability post-intervention across three dimensions: knowledge, instruction-following, and perplexity. For knowledge evaluation, we employ the comprehensive MMLU multiple-choice question-answering task (Hendrycks et al., 2020). Instruction-following is assessed using MT-Bench (Zheng et al., 2023), a challenging multi-turn benchmark that tests an LLM's ability to produce coherent, informative, and engaging responses. Finally, perplexity is measured using the widely recognized WikiText2-2 benchmark (Merity et al., 2016).